

Section : Biology

Q.1 The basic steps involved in the genetic modification of an organism

include -

- (A) Identification of DNA with desirable genes
- (B) Production and clinical trial of drugs
- (C) Introduction of the identified DNA into the host
- (D) Maintenance of introduced DNA in the host and transfer of the DNA to its progeny

Choose the most appropriate combination from the options given below.

- 1. (A), (B) and (C) only
- 2. (B), (C) and (D) only
- 3. (A), (C) and (D) only
- 4. (B) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.2 Match List-I with List-II

LIST I

LIST II

Population interaction Example

- | | |
|------------------|--------------------------------|
| (A) Parasitism | (I) Herbivores and plants |
| (B) Commensalism | (II) Human liver fluke |
| (C) Mutualism | (III) Egret and grazing cattle |
| (D) Competition | (IV) Lichens |

Choose the correct answer from the options given below:

- 1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
- 2. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)
- 3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
- 4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.3 Match List-I with List-II

List-I

List-II

Structure Relation

- | | |
|---------------|-------------------|
| (A) Thalamus | (I) Seed |
| (B) Nucellus | (II) Fruit |
| (C) Ovule | (III) Germination |
| (D) Micropyle | (IV) Perisperm |

Choose the correct answer from the options given below:

- 1. (A) - (II), (B) - (IV), (C) - (I), (D) - (III)
- 2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
- 3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
- 4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.4 Disturbance in the value of allele frequency in Hardy Weinberg equilibrium is directly proportional to the:

- 1. Disturbance in gene flow
- 2. Extent of evolutionary change in a population
- 3. Gene pool of a population
- 4. Retention of original gene in new population

Options 1. 1

2. 2

3. 3

4. 4

Q.5 Reproductive capacity of a flowering plant can be appreciated by finding out the:

(A) Size of the embryo sac in the ovule

(B) Number of eggs in the embryo sac

(C) Size of eggs in the embryo sac

(D) Number of ovules in the ovary

Choose the most appropriate answer from the options given below:

1. (A) and (C) only

2. (B) and (D) only

3. (A), (B), (C) and (D)

4. (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.6 Amniocentesis is legally used to determine---

(A) Sex of the foetus

(B) Presence of genetic disorders

(C) Massive child immunization.

(D) Survivability of the fetus

Choose the correct answer from the options given below:

1. (A), (C) and (D) only.

2. (B), (C) and (D) only.

3. (B) and (D) only.

4. (A), (B) and (C) only.

Options 1. 1

2. 2

3. 3

4. 4

Q.7 Which amino acids are present in the histone octamer that contribute to the stable packaging of DNA?

1. Methionine and proline

2. Valine and Arginine

3. Methionine and lysine

4. Lysine and arginine

Options 1. 1

2. 2

3. 3

4. 4

Q.8 Completion of the first meiotic division of the primary oocytes takes place -

1. When acrosomal secretion of the sperm enters zona pellucida of the ovum

2. Within the tertiary follicle

3. When the ovum reaches the ampullary region of the fallopian tube

4. During copulation

Options 1. 1

2. 2

3. 3

4. 4

Q.9 Which of the following are autosomal recessive gene disorders?

(A) Thalassemia

(B) Phenylketonuria

(C) Haemophilia

(D) Sickle cell anemia

Choose the correct answer from the options given below:

1. (C) and (A) only
2. (A), (B) and (D) only
3. (A) and (D) only
4. (B) and (C) only

Options 1. 1

2. 2
3. 3
4. 4

Q.10 Depending upon the immediate urgency of the need to conserve biodiversity, arrange the options given below in a descending order.

- (A) Hot spots
- (B) Sacred groves
- (C) *Ex situ* method
- (D) Bioreserves

Choose the most appropriate answer from the options given below:

1. (A), (B), (C), (D)
2. (A), (C), (B), (D)
3. (B), (A), (D), (C)
4. (C), (B), (D), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.11 Meiotic division of the secondary oocyte takes place in the:

1. Isthmus of oviduct
2. Ampulla
3. Uterus
4. Cervix

Options 1. 1

2. 2
3. 3
4. 4

Q.12 Which one of the following is an example of man-made ecosystem?

1. Agricultural field
2. Forest
3. Desert
4. River

Options 1. 1

2. 2
3. 3
4. 4

Q.13 Arrange the following stages of oogenesis in correct sequence of their formation.

- (A) Secondary oocyte
- (B) Oogonials
- (C) Ovum
- (D) Primary oocyte

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (C), (D), (B), (A)
3. (A), (C), (B), (D)
4. (B), (D), (A), (C)

Options 1. 1

2. 2
3. 3
4. 4

Q.14 Identify the incorrect example of albuminous seed.

1. Wheat
2. Castor

3. Groundnut

4. Barley

Options 1. 1

2. 2

3. 3

4. 4

Q.15 Cloning of alien DNA by the host involves -

1. Attaching at the specific sequence site (ori) in the host
2. Complementing recombinant with non recombinant fragments
3. Ligating at multiple recognition sites
4. Insertion activation technique

Options 1. 1

2. 2

3. 3

4. 4

Q.16 Choose the most appropriate option about the presence of nitrogenous bases; Cytosine, 5-methyl uracil and Uracil.

1. Cytosine only in DNA, 5-Methyl Uracil in both; DNA and RNA, Uracil only in RNA
2. Cytosine only in RNA, 5-Methyl Uracil only in DNA, Uracil in both Dna and RNA
3. Cytosine in both; DNA and RNA, 5-Methyl Uracil only in DNA and Uracil only in RNA
4. Cytosine only in RNA, 5-Methyl Uracil in both; DNA and RNA, Uracil only in DNA

Options 1. 1

2. 2

3. 3

4. 4

Q.17 Match List-I with List-II

List-I

List-II

Abbreviation Relation

- | | |
|----------|---------------------------------------|
| (A) MTP | (I) Infertility |
| (B) AIDS | (II) Artificial insemination |
| (C) IUI | (III) Legal |
| (D) ART | (IV) Careless exchange of body fluids |

Choose the correct answer from the options given below:

1. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)
2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (II), (C) - (IV), (D) - (I)

Options 1. 1

2. 2

3. 3

4. 4

Q.18 Chronologically arrange the names of scientists who contributed to the study of genetics.

- (A) Alfred Sturtevent
- (B) Gregor Mendel
- (C) Thomas Hunt Morgan
- (D) Sutton and Boveri

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (A), (C), (B), (D)
3. (B), (D), (C), (A)
4. (C), (B), (D), (A)

Options 1. 1

2. 2

3. 3

4. 4

Q.19 Baculoviruses are excellent candidates for species specific _____ application.

1. Weedicide
2. Insecticidal
3. General pest control
4. Pesticide

Options 1. 1

2. 2

3. 3

4. 4

Q.20 The composition of biogas is -

1. CH₄, H₂S and N₂
2. CH₄, H₂ and O₂
3. CH₄, H₂S and CO₂
4. CH₄, SO₂ and CO₂

Options 1. 1

2. 2

3. 3

4. 4

Q.21 Pyramid of energy is always upright because:

1. Energy flows unidirectionally
2. The number of organisms at different trophic levels do not affect energy
3. Some energy is lost when we move from one trophic level below to next trophic level above
4. Energy largely depends on the biomass at the base trophic level

Options 1. 1

2. 2

3. 3

4. 4

Q.22 Which one of the following techniques is most appropriate where the female cannot produce an ovum but can provide the environment for the development of embryo?

1. ICSI
2. AI
3. GIFT
4. IUI

Options 1. 1

2. 2

3. 3

4. 4

Q.23 Population density at time $t + 1$ is typically expressed as:

$N_{t+1} = N_t + \{B + I\} - \{D + E\}$ where $B + I$,

1. Inversely proportional to $B + I$
2. Inversely proportional to $D + E$
3. Directly proportional to E
4. Directly proportional to D

Options 1. 1

2. 2

3. 3

4. 4

Q.24 In agarose gel electrophoresis, the separated DNA fragments can be visualised by staining the DNA with a compound, known as

1. Ethidium bromate
2. Ethidium chloride
3. Ethidium bromide

4. Ethidium bromine

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.25 Find out the odd one out among the following structures in relation to evolution?

- 1. Sweet potato and potato
- 2. Penguin and Dolphin flippers
- 3. Mammals and Octopus Eyes
- 4. Thorns of *Bougainvillea* and tendrils of *Cucurbita*

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.26 First meiotic division of primary follicles completes in the

- 1. Fetal ovary
- 2. Tertiary follicle
- 3. Graafian follicle
- 4. Before puberty

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.27 Which one of the following is not a cause of loss of biodiversity?

- 1. Afforestation
- 2. Habitat loss
- 3. Alien species invasions
- 4. Co-extinction

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.28 With reference to the microorganisms involved in production, identify the odd combination.

- 1. Bread - house hold product
- 2. Penicillin - antibiotic
- 3. Lactic acid - industrial product
- 4. Biogas - energy source

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.29 Host bacterium is made competent to accept DNA by treating it with:

- 1. Prolonged heat shock
- 2. Short exposure to cold temperature
- 3. Divalent cation
- 4. Monovalent anion

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.30 Bioactive molecule, cyclosporin A, that is used as an immunosuppressive agent in organ-transplant patients, is produced by the fungus -

- 1. *Glomus*
- 2. *Monascus purpureus*

3. *Streptococcus*
4. *Trichoderma polysporum*

Options 1. 1

2. 2
3. 3
4. 4

Q.31 The density of population is decreased by the processes

- (A) Natality
(B) Emigration
(C) Mortality
(D) Immigration

Choose the most appropriate combination from the options given below.

1. (A) and (B) only
2. (B) and (C) only
3. (A), (B) and (D) only
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.32 Exponential growth rate of a population is graphically represented by -

1. Bell shaped curve
2. Sigmoid curve
3. C-shaped curve
4. J-shaped curve

Options 1. 1

2. 2
3. 3
4. 4

Q.33 Arrange the evolutionary human species in their occurrence from recent to the oldest order.

- (A) *Australopithecus*
(B) *Homo erectus*
(C) *Homo sapiens*
(D) Neanderthal man

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (C), (D), (B), (A)
3. (B), (A), (D), (C)
4. (C), (B), (D), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.34 Match List-I with List-II

List-I

List-II

Term

Chromosome formulation

- (A) Diploid (I) $2n-1$
(B) Aneuploidy (II) $4n$
(C) Haploid (III) n
(D) Polyploidy (IV) $2n$

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (IV), (B) - (I), (C) - (III), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.35 In plants, competency of the host to take up the recombinant DNA is enabled by using _____ technique.

1. Insertional inactivation
2. Micro injection
3. Biolistics
4. Gel electrophoresis

Options 1. 1

2. 2

3. 3

4. 4

Q.36 Match List-I with List-II

List-I

List-II

Term

Feature

(A) Multiple alleles

(I) Variation in DNA

(B) Mendelian disorder

(II) Non segregation of chromatids during cell division

(C) Chromosomal disorder

(III) Pedigree analysis

(D) Mutation

(IV) Population study

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.37 Minimum percentage of global biodiversity in major invertebrate and plant taxa, is seen in:

1. Molluscs and algae
2. Crustaceans and lichens
3. Crustaceans and ferns
4. Insects and mosses

Options 1. 1

2. 2

3. 3

4. 4

Q.38 Arrange the following wall layers of microsporangium starting from outer to inner layer.

- (A) Tapetum
- (B) Endothecium
- (C) Middle layers
- (D) Epidermis

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (D), (B), (C), (A)
3. (B), (A), (D), (C)
4. (A), (D), (B), (C)

Options 1. 1

2. 2

3. 3

4. 4

Q.39 Statins are produced by the yeast -

1. *Monascus purpureus*

2. *Trichoderma polysporum*
3. *Acetobacter aceti*
4. *Saccharomyces cerevisiae*

Options 1. 1

2. 2
3. 3
4. 4

Q.40 The typical test cross is performed between -

1. An organism showing a dominant phenotype and the recessive parent
2. An organism showing a dominant phenotype and the dominant parent
3. An organism showing a dominant genotype and the dominant parent
4. An organism showing a recessive phenotype and the dominant parent

Options 1. 1

2. 2
3. 3
4. 4

Q.41 Read the passage carefully to answer the questions given below
Some strains of *Bacillus thuringiensis* produce proteins that kill certain insects such as lepidopterans (tobacco budworm, armyworm), coleopterans (beetles) and dipterans (flies, mosquitoes). In *Bacillus thuringiensis* the toxin is coded by a gene *cryI*Ac named *cry*. There are a number of them, for example, the proteins encoded by the genes *cryI*Ac and *cryII*Ab control the cotton bollworms, that of *cryI*Ab controls corn borer. Using conventional methods of diagnosis (serum and urine analysis, etc.) early detection is not possible. Recombinant DNA technology, Polymerase Chain Reaction (PCR) and Enzyme Linked Immuno-sorbent Assay (ELISA) are some of the techniques that serve the purpose of early diagnosis. The first clinical gene therapy was given in 1990 to a 4-year old girl with adenosine deaminase (ADA) deficiency. This enzyme is crucial for the immune system to function. The disorder is caused due to the deletion of the gene for adenosine deaminase. RNAi takes place in all eukaryotic organisms as a method of cellular defense. This method involves silencing of a specific mRNA due to a complementary dsRNA molecule that binds to and prevents translation of the mRNA.

Some strains of *Bacillus thuringiensis* produce proteins that do not kill ceratin organisms -

1. Crustaceans
2. Dipterans
3. Lepidopterans
4. Coleopterans

Options 1. 1

2. 2
3. 3
4. 4

Q.42 Read the passage carefully to answer the questions given below
Some strains of *Bacillus thuringiensis* produce proteins that kill certain insects such as lepidopterans (tobacco budworm, armyworm), coleopterans (beetles) and dipterans (flies, mosquitoes). In *Bacillus thuringiensis* the toxin is coded by a gene *cryI*Ac named *cry*. There are a number of them, for example, the proteins encoded by the genes *cryI*Ac and *cryII*Ab control the cotton bollworms, that of *cryI*Ab controls corn borer. Using conventional methods of diagnosis (serum and urine analysis, etc.) early detection is not possible. Recombinant DNA technology, Polymerase Chain Reaction (PCR) and Enzyme Linked Immuno-sorbent Assay (ELISA) are some of the techniques that serve the purpose of early diagnosis. The first clinical gene therapy was given in 1990 to a 4-year old girl with adenosine deaminase (ADA) deficiency. This enzyme is crucial for the immune system to function. The disorder is caused due to the deletion of the gene for adenosine deaminase. RNAi takes place in all eukaryotic organisms as a method of cellular defense. This method involves silencing of a specific mRNA due to a complementary dsRNA molecule that binds to and prevents translation of the mRNA.

RNAi involves silencing of a specific mRNA due to a complementary _____ molecule that binds to and prevents translation

of the mRNA.

1. dsRNA
2. ssRNA
3. dsDNA
4. ssDNA

Options 1. 1

2. 2
3. 3
4. 4

Q.43 Read the passage carefully to answer the questions given below
Some strains of *Bacillus thuringiensis* produce proteins that kill certain insects such as lepidopterans (tobacco budworm, armyworm), coleopterans (beetles) and dipterans (flies, mosquitoes). In *Bacillus thuringiensis* the toxin is coded by a gene cryI_{Ac} named cry. There are a number of them, for example, the proteins encoded by the genes cryI_{Ac} and cryII_{Ab} control the cotton bollworms, that of cryI_{Ab} controls corn borer. Using conventional methods of diagnosis (serum and urine analysis, etc.) early detection is not possible. Recombinant DNA technology, Polymerase Chain Reaction (PCR) and Enzyme Linked Immuno-sorbent Assay (ELISA) are some of the techniques that serve the purpose of early diagnosis. The first clinical gene therapy was given in 1990 to a 4-year old girl with adenosine deaminase (ADA) deficiency. This enzyme is crucial for the immune system to function. The disorder is caused due to the deletion of the gene for adenosine deaminase. RNAi takes place in all eukaryotic organisms as a method of cellular defense. This method involves silencing of a specific mRNA due to a complementary dsRNA molecule that binds to and prevents translation of the mRNA.

Which of the following technique is not used for early diagnosis of diseases?

1. Recombinant DNA technology
2. Polymerase chain reaction
3. Serum analysis
4. Enzyme linked immuno-sorbent assay

Options 1. 1

2. 2
3. 3
4. 4

Q.44 Read the passage carefully to answer the questions given below
Some strains of *Bacillus thuringiensis* produce proteins that kill certain insects such as lepidopterans (tobacco budworm, armyworm), coleopterans (beetles) and dipterans (flies, mosquitoes). In *Bacillus thuringiensis* the toxin is coded by a gene cryI_{Ac} named cry. There are a number of them, for example, the proteins encoded by the genes cryI_{Ac} and cryII_{Ab} control the cotton bollworms, that of cryI_{Ab} controls corn borer. Using conventional methods of diagnosis (serum and urine analysis, etc.) early detection is not possible. Recombinant DNA technology, Polymerase Chain Reaction (PCR) and Enzyme Linked Immuno-sorbent Assay (ELISA) are some of the techniques that serve the purpose of early diagnosis. The first clinical gene therapy was given in 1990 to a 4-year old girl with adenosine deaminase (ADA) deficiency. This enzyme is crucial for the immune system to function. The disorder is caused due to the deletion of the gene for adenosine deaminase. RNAi takes place in all eukaryotic organisms as a method of cellular defense. This method involves silencing of a specific mRNA due to a complementary dsRNA molecule that binds to and prevents translation of the mRNA.

The first clinical gene therapy was given in 1990 to a 4-year old girl with adenosine deaminase (ADA) deficiency.

This disorder is caused due to the deletion of the gene for -

1. Adenosine decarboxylase
2. Adenosine dehydrogenase
3. Adenosine deaminase
4. Adenosine deacetylase

Options 1. 1

2. 2
3. 3

Q.45 Read the passage carefully to answer the questions given below
Some strains of *Bacillus thuringiensis* produce proteins that kill certain insects such as lepidopterans (tobacco budworm, armyworm), coleopterans (beetles) and dipterans (flies, mosquitoes). In *Bacillus thuringiensis* the toxin is coded by a gene *cryIAC* named *cry*. There are a number of them, for example, the proteins encoded by the genes *cryIAC* and *cryIIAb* control the cotton bollworms, that of *cryIAb* controls corn borer. Using conventional methods of diagnosis (serum and urine analysis, etc.) early detection is not possible. Recombinant DNA technology, Polymerase Chain Reaction (PCR) and Enzyme Linked Immuno-sorbent Assay (ELISA) are some of the techniques that serve the purpose of early diagnosis. The first clinical gene therapy was given in 1990 to a 4-year old girl with adenosine deaminase (ADA) deficiency. This enzyme is crucial for the immune system to function. The disorder is caused due to the deletion of the gene for adenosine deaminase. RNAi takes place in all eukaryotic organisms as a method of cellular defense. This method involves silencing of a specific mRNA due to a complementary dsRNA molecule that binds to and prevents translation of the mRNA.

Which of the following Bt toxin gene controls the corn borer?

1. *CryIAC*
2. *CryIIAC*
3. *CryIAb*
4. *CryIIAb*

Options 1. 1

2. 2
3. 3
4. 4

Q.46 Read the passage carefully and answer the questions given along
Our immune system plays the major role in preventing diseases when we are exposed to disease-causing agents. Immunity is of two types: (i) Innate immunity and (ii) Acquired immunity. The human immune system consists of lymphoid organs, tissues, cells and soluble molecules like antibodies. The immune system also plays an important role in allergic reactions, autoimmune diseases and organ transplantation. The primary lymphoid organs are bone marrow and thymus. The secondary lymphoid organs are spleen, lymph nodes, tonsils, Peyer's patches of small intestine and appendix. There is lymphoid tissue also located within the lining of the major tracts (respiratory, digestive and urogenital tracts) called mucosa associated lymphoid tissue (MALT). It is important to note that HIV/AIDS is not spread by mere touch or physical contact; it spreads only through body fluids. AIDS has no cure, prevention is the best option. Adolescence is accompanied by several biological and behavioural changes and surveys and statistics show that use of drugs and alcohol has been on the rise especially among the youth. Factors associated with drug and alcohol abuse among adolescents are a child's natural curiosity, stress, the perception among youth that it is 'cool' or progressive to smoke, media exposure and an unstable or unsupportive family structures and peer pressure. It is best to identify the situations that may push an adolescent towards use of drugs or alcohol, and to take remedial measures well in time.

Active immunization is most suitable method for:

1. Exposure to snake venom
2. Natural infection
3. Deadly microbe infection
4. Protection of fetus in the womb

Options 1. 1

2. 2
3. 3
4. 4

Q.47 Read the passage carefully and answer the questions given along
Our immune system plays the major role in preventing diseases when we are exposed to disease-causing agents. Immunity is of two types: (i) Innate immunity and (ii) Acquired immunity. The human immune system consists of lymphoid organs, tissues, cells and soluble molecules like antibodies. The immune system also plays an important role in allergic reactions, auto-

immune diseases and organ transplantation. The primary lymphoid organs are bone marrow and thymus. The secondary lymphoid organs are spleen, lymph nodes, tonsils, Peyer's patches of small intestine and appendix. There is lymphoid tissue also located within the lining of the major tracts (respiratory, digestive and urogenital tracts) called mucosa associated lymphoid tissue (MALT). It is important to note that HIV/AIDS is not spread by mere touch or physical contact; it spreads only through body fluids. AIDS has no cure, prevention is the best option. Adolescence is accompanied by several biological and behavioural changes and surveys and statistics show that use of drugs and alcohol has been on the rise especially among the youth. Factors associated with drug and alcohol abuse among adolescents are a child's natural curiosity, stress, the perception among youth that it is 'cool' or progressive to smoke, media exposure and an unstable or unsupportive family structures and peer pressure. It is best to identify the situations that may push an adolescent towards use of drugs or alcohol, and to take remedial measures well in time.

Mucosa associated lymphoid tissue is absent in the

1. Respiratory tract
2. Urogenital system
3. Digestive tract
4. Skeletal system

Options 1. 1

2. 2
3. 3
4. 4

Q.48 Read the passage carefully and answer the questions given along
Our immune system plays the major role in preventing diseases when we are exposed to disease-causing agents. Immunity is of two types: (i) Innate immunity and (ii) Acquired immunity. The human immune system consists of lymphoid organs, tissues, cells and soluble molecules like antibodies. The immune system also plays an important role in allergic reactions, auto-immune diseases and organ transplantation. The primary lymphoid organs are bone marrow and thymus. The secondary lymphoid organs are spleen, lymph nodes, tonsils, Peyer's patches of small intestine and appendix. There is lymphoid tissue also located within the lining of the major tracts (respiratory, digestive and urogenital tracts) called mucosa associated lymphoid tissue (MALT). It is important to note that HIV/AIDS is not spread by mere touch or physical contact; it spreads only through body fluids. AIDS has no cure, prevention is the best option. Adolescence is accompanied by several biological and behavioural changes and surveys and statistics show that use of drugs and alcohol has been on the rise especially among the youth. Factors associated with drug and alcohol abuse among adolescents are a child's natural curiosity, stress, the perception among youth that it is 'cool' or progressive to smoke, media exposure and an unstable or unsupportive family structures and peer pressure. It is best to identify the situations that may push an adolescent towards use of drugs or alcohol, and to take remedial measures well in time.

Which one of the following can never be over estimated when it comes to the complete well-being of adolescents?

1. Drugs
2. Environment
3. Biology
4. Alcohol

Options 1. 1

2. 2
3. 3
4. 4

Q.49 Read the passage carefully and answer the questions given along
Our immune system plays the major role in preventing diseases when we are exposed to disease-causing agents. Immunity is of two types: (i) Innate immunity and (ii) Acquired immunity. The human immune system consists of lymphoid organs, tissues, cells and soluble molecules like antibodies. The immune system also plays an important role in allergic reactions, auto-immune diseases and organ transplantation. The primary lymphoid organs are bone marrow and thymus. The secondary lymphoid organs are spleen,

lymph nodes, tonsils, Peyer's patches of small intestine and appendix. There is lymphoid tissue also located within the lining of the major tracts (respiratory, digestive and urogenital tracts) called mucosa associated lymphoid tissue (MALT). It is important to note that HIV/AIDS is not spread by mere touch or physical contact; it spreads only through body fluids. AIDS has no cure, prevention is the best option. Adolescence is accompanied by several biological and behavioural changes and surveys and statistics show that use of drugs and alcohol has been on the rise especially among the youth. Factors associated with drug and alcohol abuse among adolescents are a child's natural curiosity, stress, the perception among youth that it is 'cool' or progressive to smoke, media exposure and an unstable or unsupportive family structures and peer pressure. It is best to identify the situations that may push an adolescent towards use of drugs or alcohol, and to take remedial measures well in time.

Find the incorrect statement about human health and disease.

1. Allergens are antigens.
2. Most pathogens are parasites.
3. AIDS is a congenital disease.
4. Bone marrow and thymus are primary lymphoid organs.

Options 1. 1

2. 2
3. 3
4. 4

Q.50 Read the passage carefully and answer the questions given along

Our immune system plays the major role in preventing diseases when we are exposed to disease-causing agents. Immunity is of two types: (i) Innate immunity and (ii) Acquired immunity. The human immune system consists of lymphoid organs, tissues, cells and soluble molecules like antibodies. The immune system also plays an important role in allergic reactions, auto-immune diseases and organ transplantation. The primary lymphoid organs are bone marrow and thymus. The secondary lymphoid organs are spleen, lymph nodes, tonsils, Peyer's patches of small intestine and appendix. There is lymphoid tissue also located within the lining of the major tracts (respiratory, digestive and urogenital tracts) called mucosa associated lymphoid tissue (MALT). It is important to note that HIV/AIDS is not spread by mere touch or physical contact; it spreads only through body fluids. AIDS has no cure, prevention is the best option. Adolescence is accompanied by several biological and behavioural changes and surveys and statistics show that use of drugs and alcohol has been on the rise especially among the youth. Factors associated with drug and alcohol abuse among adolescents are a child's natural curiosity, stress, the perception among youth that it is 'cool' or progressive to smoke, media exposure and an unstable or unsupportive family structures and peer pressure. It is best to identify the situations that may push an adolescent towards use of drugs or alcohol, and to take remedial measures well in time.

Which one of the following structures is not an example of a cellular barrier in innate immunity?

1. Protein interferon
2. PMNL neutrophils
3. Lymphocytes
4. Macrophages

Options 1. 1

2. 2
3. 3
4. 4